

Broccoli sprouts reduce oxidative stress in type 2 diabetes: a randomized double-blind clinical trial.

BACKGROUND/OBJECTIVES: In vitro and animal studies have reported that young broccoli sprouts improve oxidative stress status in diabetic condition. The objective of this double-blind, placebo-controlled, randomized clinical trial was to investigate the effects of broccoli sprouts powder (BSP) on some oxidative stress parameters in type 2 diabetes patients. **SUBJECTS/METHODS:** A total of 81 patients with type 2 diabetes were randomly assigned to one of three treatment groups for 4 weeks. The groups received either 10 g/d BSP (n=27), 5 g/d BSP (n=29) or placebo (n=25). Serum total antioxidant capacity (TAC), total oxidant status (TOS), oxidative stress index (OSI), malondialdehyde (MDA) and oxidized low density lipoprotein (LDL) cholesterol were measured at baseline and at 4 weeks after treatment. **RESULTS:** In all, 63 patients in three groups were included in the analysis: 10 g/d BSP (n=21), 5 g/d (n=22) and placebo (n=20). After 4 weeks, consumption of BSP resulted in significant decrease in MDA ($P=0.001$ for treatment effect), oxidized low density lipoprotein cholesterol ($P=0.03$ for treatment effect), OSI ($P=0.001$ for treatment effect) and significant increase in TAC ($P=0.001$ for treatment effect). No effects were found on TOS. **CONCLUSION:** BSP had favorable effects on oxidative stress status in type 2 diabetes patients.